

# Day 5 — Technical Trade-Offs & Constraints (TCD Assembly + Review) · Facilitator Guide

Companion to `labs/day-05-tradeoffs-tcd.md` . Facilitator-only — do not hand to participants.

## Framing note

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Today's job is to name the **top 5 trade-offs** (TCD §5), build the **stakeholder sign-off matrix** (§6), assemble the integrated TCD, peer-review it, and ship it as the sibling artifact to the PRD.

## Activity 1 — Surface the Trade-Offs

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### Cues:

- Strawman alternatives ("the other option was obviously worse") signal under-thinking. Push for a real Option B.
- All-one-category lists usually missed real tensions; mix coupling, consistency, latency, speed, or generality.

## Activity 2 — Write the Top 5 Trade-Offs

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### Cues:

- A trade-off that ends "we chose A because A is better" with no actual reasoning — push back. Force the cost and trigger.
- Watch for "we accept complexity" — that's vague. What complexity? What does it cost in maintenance, debugging, on-call?
- Trade-offs that align suspiciously well with personal preference are a smell. The reasoning should come from the system, not from "I like X better."

## Activity 3 — Stakeholder Sign-Off Matrix

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### Cues:

- "TBD" stakeholders signal an unowned constraint that will slip. Force a real name even if it's the architect placeholder.
- Matrices that overstate status ("Approved" without evidence) become Week 6 surprises; coach toward honesty.

## Activity 4 — Integration + Cross-Review + Sign-Off

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### Cues:

- Watch for "Approved" without an honest gap list. The rubric should set the bar, not the triad's confidence.
- Reviewers who score generously should be calibrated against the rubric mid-block.

## Facilitator wrap (15 min, end of day)

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- Read aloud one trade-off from each triad's TCD that the cohort should learn from.
- Surface the **most common pattern of trade-off avoidance** the cohort exhibited (this is a Week 5 + 6 coaching theme).
- Preview Week 5: TCD becomes the spine. Data modeling will reference §2; performance baselines will reference §4; APIs will reference §3 (auth) and §4 (rate limits).

## Facilitator reflection prompts (end of week)

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- Which triad's TCD reads like an architect could pick it up? They are the Week-5 positive example.
- Which triad's stakeholder matrix is the weakest? Coach individually before Week 5.
- Did any triad over-specify and accidentally make architectural decisions? Coach toward "surface trade-offs, don't decide."
- AI use this week: was the discipline maintained? If not, surface for the cohort.